ABID ARHAM

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SKILLS SUMMARY

- **Languages:** TypeScript, JavaScript, C++, Java, PHP, SQL, HTML, CSS
- **Frontend:** React, Next.js (App Router), Redux Toolkit, TanStack Query, Zustand, Tailwind CSS, shadcn/ui
- **Backend:** Node.js, Express, REST APIs, JWT & Passport auth, RBAC, Prisma ORM, Mongoose, Zod, Stripe
- **Databases:** PostgreSQL, MySQL, MongoDB; both raw SQL and ORM-based data access
- **Tools & Platforms:** Git, GitHub, Docker, Postman, Vercel, Render, Cloudinary, Redis (Upstash)
- **Fundamentals:** Data structures & algorithms, OOP, design patterns, software architecture, problem solving
- **Soft Skills:** Client communication, requirement gathering, independent delivery, technical writing

PROJECTS

Quad — University Management System | Live | Frontend | API **September 2026 – October 2026**

Next.js 16, TypeScript, TanStack Query, Express 5, Prisma, PostgreSQL, Stripe, Redis, Cloudinary

- Built a three-role platform where students register for courses and pay tuition through Stripe Checkout, instructors grade their own sections, and the registrar manages departments, semesters, sections, and the course catalogue.
- Secured authentication by keeping JWTs in httpOnly cookies behind a backend-for-frontend proxy, with route-level role checks in Next.js middleware and silent, single-flight token refresh; access tokens rotate every 15 minutes and refresh tokens are stored hashed.
- Server-prefetched dashboards with TanStack Query hydration to eliminate duplicate fetching, and added optimistic grading and enrollment updates plus a 4-step course wizard whose draft survives a page refresh.
- Implemented transaction-safe enrollment with capacity checks and prerequisite cycle detection, admin analytics built from an audit log with Recharts, and a printable transcript handling term GPA, cumulative GPA, and retakes.

White.H — Clothing & Beauty E-Commerce Store | Live | Client project (private repository) **Live in production**

React, Redux Toolkit, TypeScript, Express, MongoDB Atlas, Docker, Resend

- Delivered a full-stack storefront for a Bangladeshi brand as a paid client project: React frontend on Vercel, Express REST API on Render, and MongoDB Atlas — my first project running in production with real customers.
- Designed a manual bKash/Nagad payment flow with an admin verification queue, duplicate transaction-ID flagging, and automated Resend email alerts, working around the lack of a local card-payment gateway.
- Hardened the API with JWT access and refresh tokens, bcrypt hashing, role-based admin routes, Zod validation, Helmet, and rate-limited auth endpoints; containerized the service with a multi-stage Dockerfile.

FixItNow — Home-Services Marketplace | Live | Repository **August 2026 – October 2026**

Next.js 16, React 19, TypeScript, Tailwind CSS 4, Express, Prisma, PostgreSQL, Claude API

- Built a three-sided marketplace where customers book vetted technicians for plumbing, electrical, cleaning, and internet work, technicians manage jobs and availability, and admins run the platform.
- Integrated an AI chat assistant and an admin briefing powered by the Claude API, with a graceful fallback to answers computed from live data when no API key is configured.
- Added Google and Facebook OAuth sign-in on top of JWT auth, and diagnosed and fixed an auth middleware bug, a per-user caching bug, and Suspense prerendering errors.

Aegis — Guard-Railed Agentic Personal-Finance Advice (Research) | Repository **September 2026**

TypeScript, Node.js — zero-dependency reference implementation

- Designed a four-layer mediation architecture that moves safety enforcement out of the LLM prompt into a separate policy layer every request must pass through; the model sees opaque handles instead of account identifiers and may only call allowlisted tools.
- Authored the research paper and a zero-dependency TypeScript reference prototype; in ablation testing, mediation blocked 15/15 policy-violating prompts (0/15 blocked with mediation disabled) with 0/6 false refusals on benign controls.
- Measured a p95 local pipeline latency of 0.061 ms and kept the design extensible — adding a new capability takes 2 files and 0 changes to enforcement code.

DevPulse — Issue & Feature Tracker REST API | Repository **June 2026**

Node.js, TypeScript, Express, PostgreSQL (native pg driver, raw SQL only)

- Built a collaborative issue tracker API for software teams with JWT and bcrypt authentication and role-based access control across contributor, maintainer, and admin roles.
- Wrote every query as raw parameterized SQL with no ORM or query builder, working directly with joins, transactions, and indexing rather than abstracting them away.

NextCart — Multi-Vendor E-Commerce Platform | Repository **May 2026 – July 2026**

Core PHP (MVC-style), MySQL, raw mysqli, plain CSS — no framework

- Built a four-portal marketplace — customer, seller, delivery, and admin — running against one shared database with unified role-based login; personally owned the admin portal in a team project.
- Implemented the platform without a framework or ORM, hand-rolling MVC-style controllers, models, and views per portal to work directly with raw SQL and the request lifecycle.

Wanderlust — Travel Listing Application | Live | Repository **March 2026**

Node.js, Express, EJS, MongoDB, Mongoose, Passport, Cloudinary, Mapbox

- Built a full-stack travel-listing app where users create, edit, and review stays, with owner-only edit and delete enforced on both listings and reviews.
- Handled image uploads through Multer and Cloudinary and added Mapbox geocoding with interactive map previews, on top of Passport local-strategy auth with session-based flash messaging.

EDUCATION

American International University-Bangladesh (AIUB) **Dhaka, Bangladesh**

B.Sc. in Computer Science and Engineering; CGPA: 3.78 / 4.00 **Expected April 2027**

Square High School & College, Pabna **Pabna, Bangladesh**

Higher Secondary Certificate (HSC), Science — GPA 5.00 / 5.00 **2021**

EXPERIENCE & ACTIVITIES

Participant | AIUB Programming Contest **March 2024**

- Solved three problems in C++ under timed contest conditions.

IT Support Specialist Trainee | SEIP (Skills for Employment Investment Program), Pabna **September 2022 – December 2022**

- Completed a government-funded training program in computer assembly, MS Office, and hardware and software troubleshooting.

Founding Member | Pabna Students Association **August 2020 – January 2021**

- Helped organize COVID-19 emergency relief distribution in the local community.